ALBION CARE NETWORK COMPANY

INTRODUCTION

Albion Care Network is one of the UK's leading private healthcare providers, operating a nationwide network of hospitals and clinics. Managing bed availability efficiently is essential to delivering high-quality patient care and maintaining operational performance. This project analyses hospital admissions, bed occupancy, emergency admissions, and elective surgeries to identify trends that support better capacity planning and informed decision-making. The insights gained can help improve resource allocation, patient flow, and overall hospital efficiency.

CHALLENGES

Hospital bed management is often reactive, leading to bed shortages during periods of high demand.

- Admissions, staffing, and bed management data are stored in separate systems, making planning difficult.
- Emergency and elective admissions fluctuate throughout the year, creating unpredictable demand for hospital beds.
- Without predictive insights, hospitals struggle to allocate beds and staff efficiently, affecting patient care and operational performance.

OBJECTIVES

Analyze historical hospital data to identify trends and seasonal patterns in admissions and bed occupancy.

- Examine the relationship between emergency admissions, elective surgeries, staffing levels, and bed occupancy.
- Support hospital capacity planning by forecasting patient demand and bed occupancy.

- Develop an interactive Power BI dashboard to provide clear, data-driven insights for decision-makers.
- Improve operational efficiency by enabling better resource allocation, staffing decisions, and proactive capacity management.

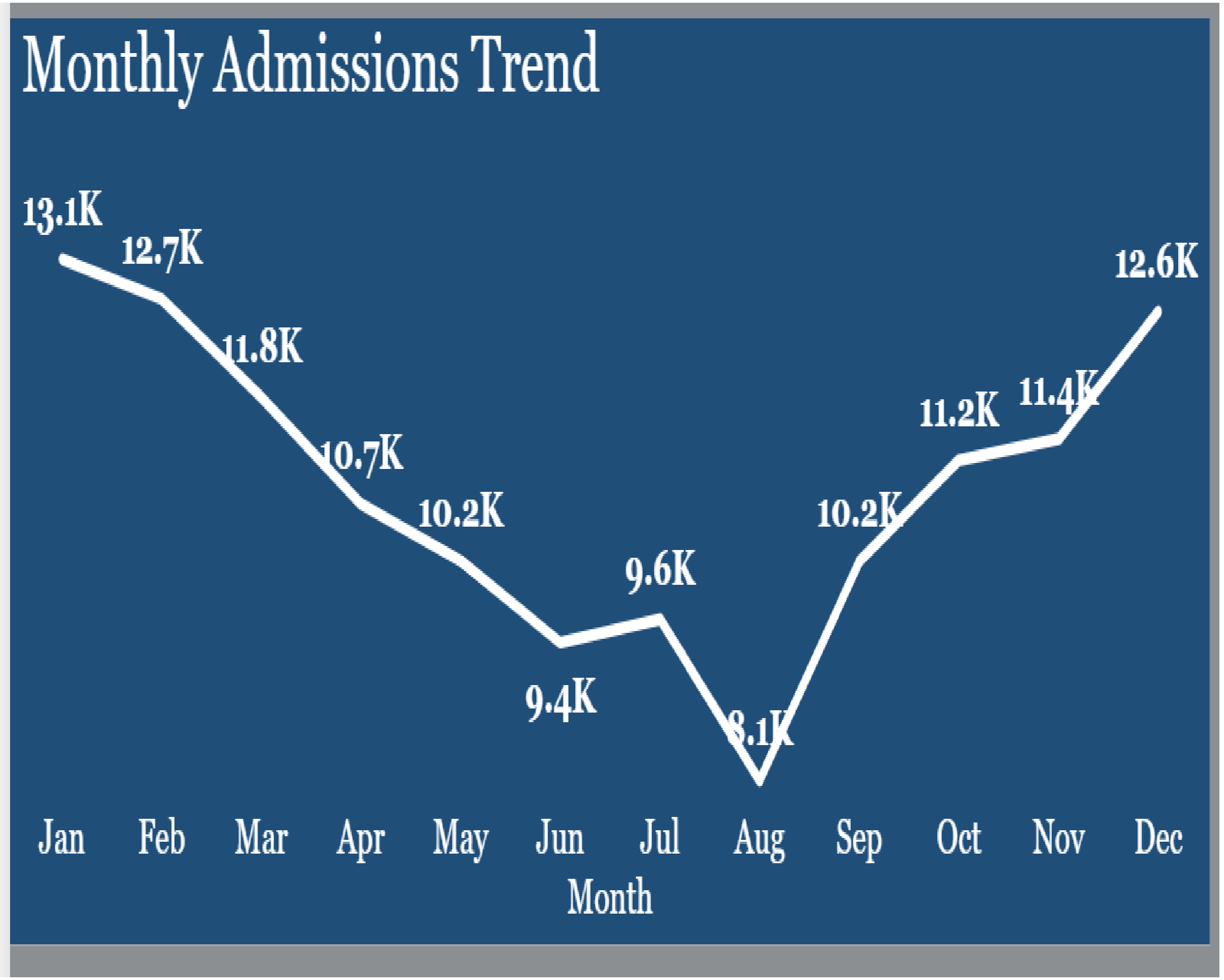
KEY METRIC

- The hospital processed **131,000** admissions during the study period.
- Average bed occupancy remained at **74%,** indicating relatively high utilization.
- **80% of admissions were emergency** cases, making emergency demand the largest contributor to hospital capacity pressure

SEASONALITY DEMAND PATTERNS

INSIGHTS

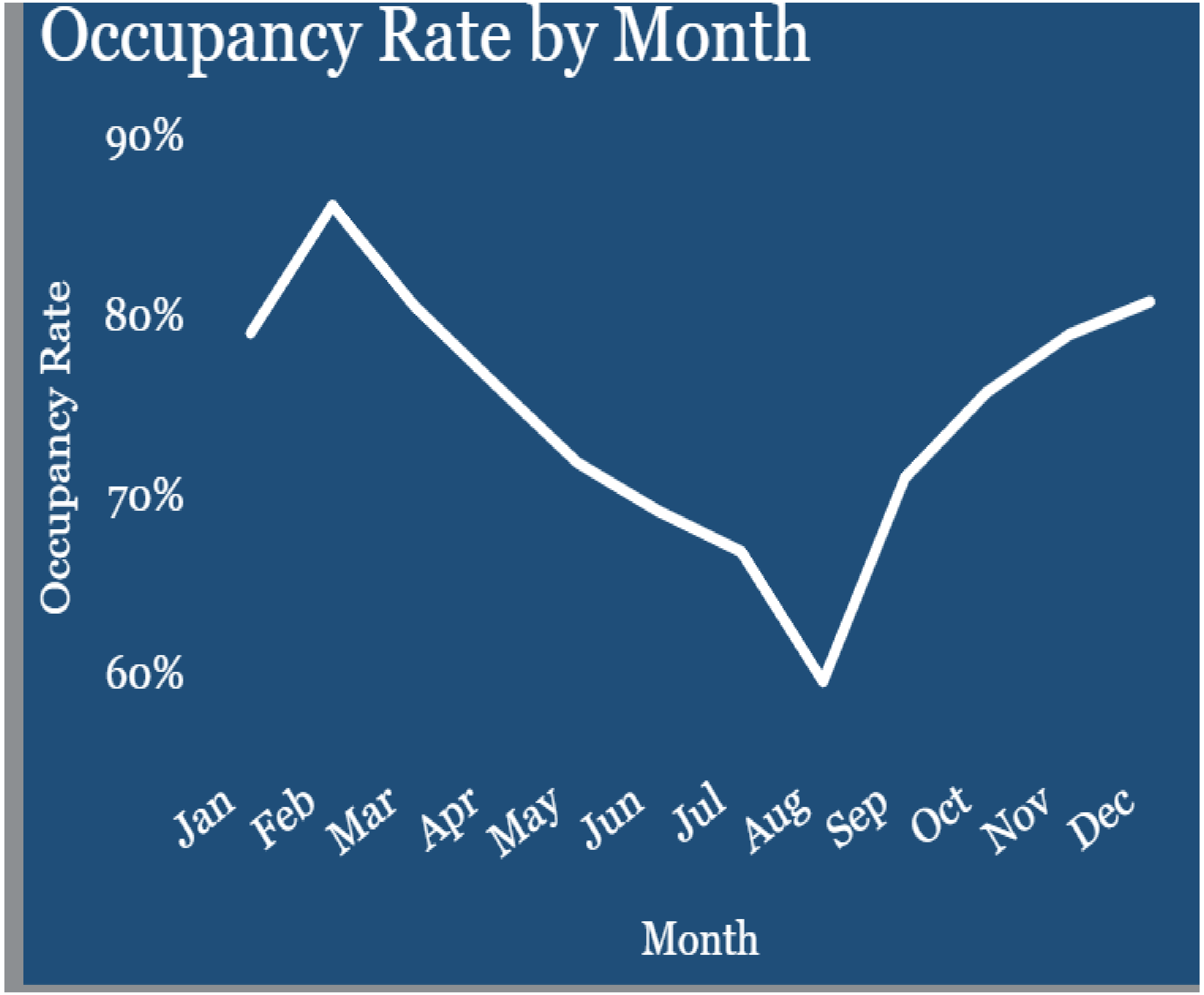
- Hospital admissions show a clear seasonal trend, declining steadily from **13.1K** in January to **12.6K** in December. This suggests that patient volumes during the final quarter by ensuring sufficient bed capacity, staffing levels and resources to maintain quality of care



SESONALITY DEMAND

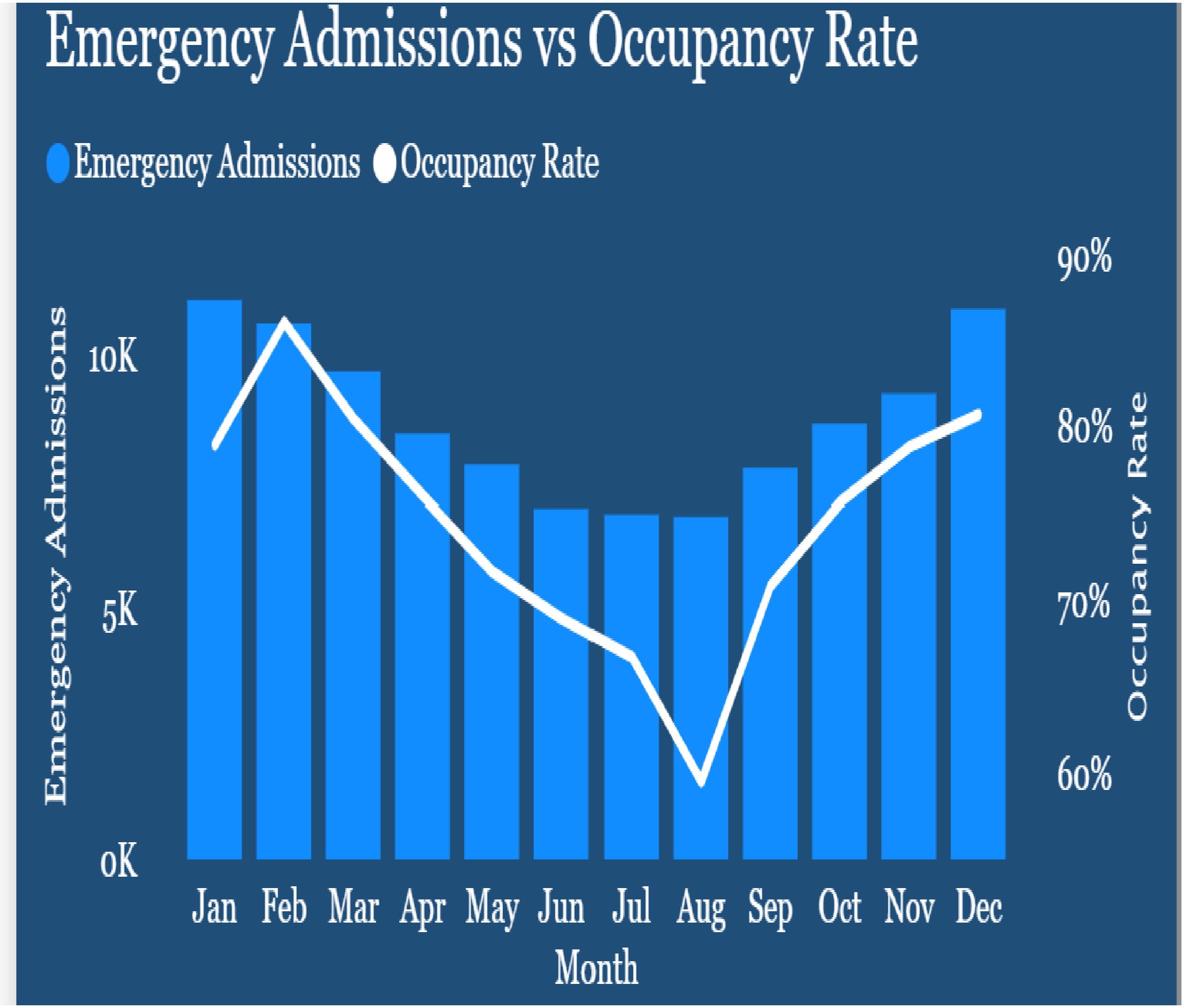
INSIGHTS

- Bed occupancy follows a clear seasonal patterns, peaking early and later in the year while reaching its lowest point in August



IMPACT ON EMERGENCY ADMISSION & BED OCCUPANCY

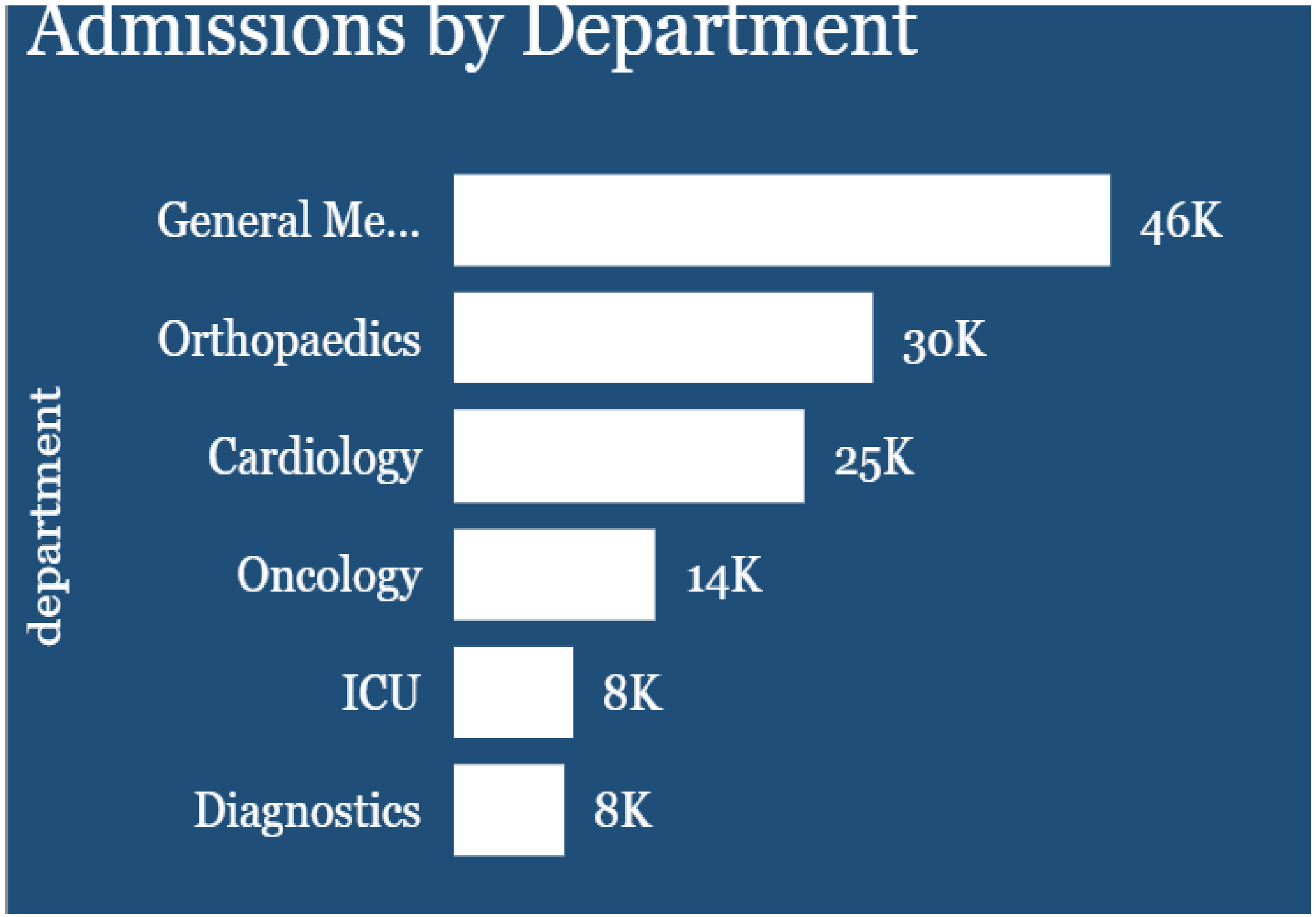
The chart shows that emergency admissions and bed occupancy move in a similar direction over the year. When emergency admissions decrease, occupancy also falls, and when emergency admissions increase towards the end of the year, occupancy rises. This suggests that emergency admissions are an important driver of bed occupancy, although they are not the only factor affecting hospital capacity



ADMISSION BY DEPARTMENT

INSIGHT

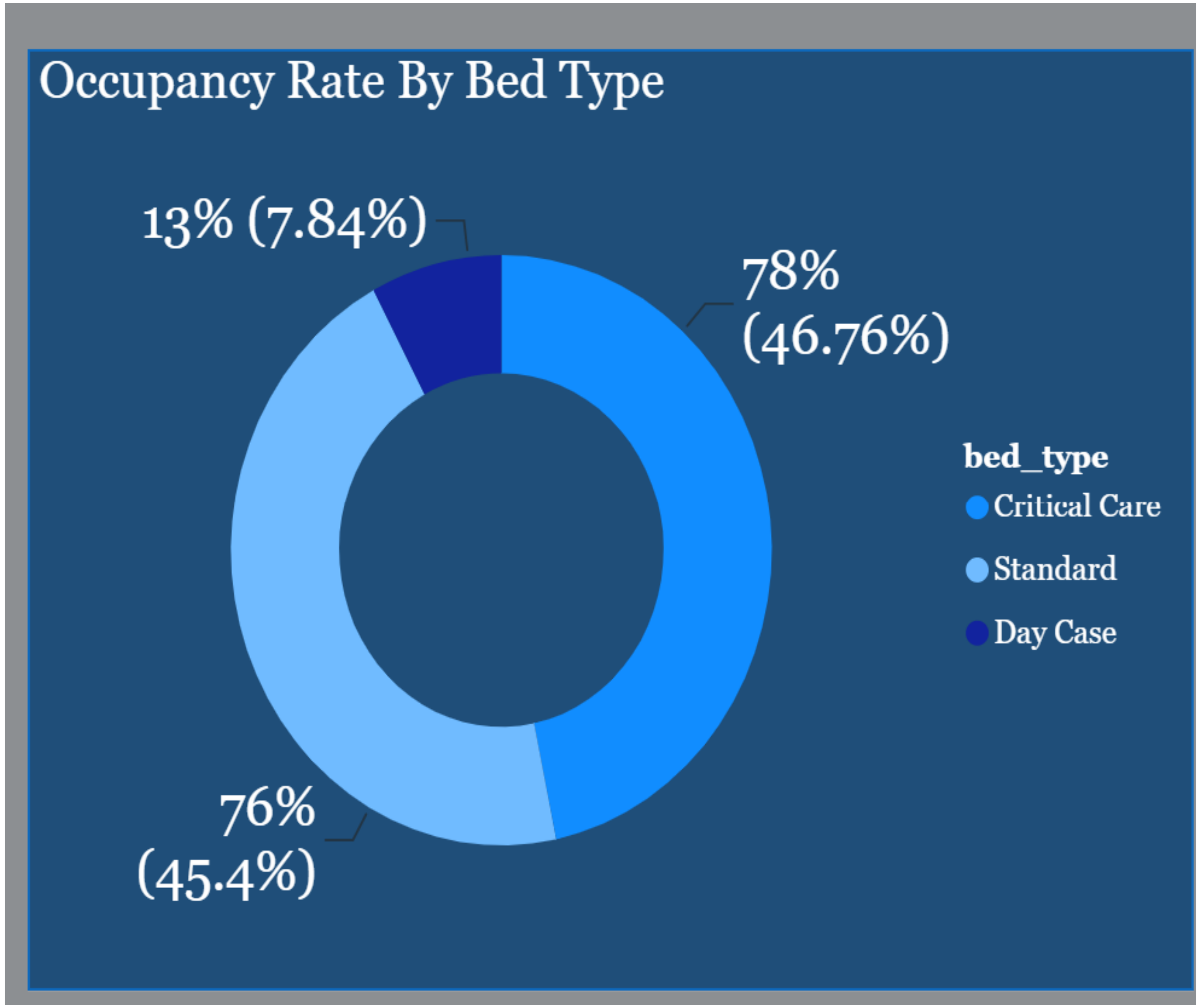
- Patient demand is highly concentrated in General Medicine **(46K),** followed by Orthopaedics **(30K)** and Cardiology **(25K),** highlighting these departments as the hospital's key service areas.



BED TYPE VS OCCUPANCY

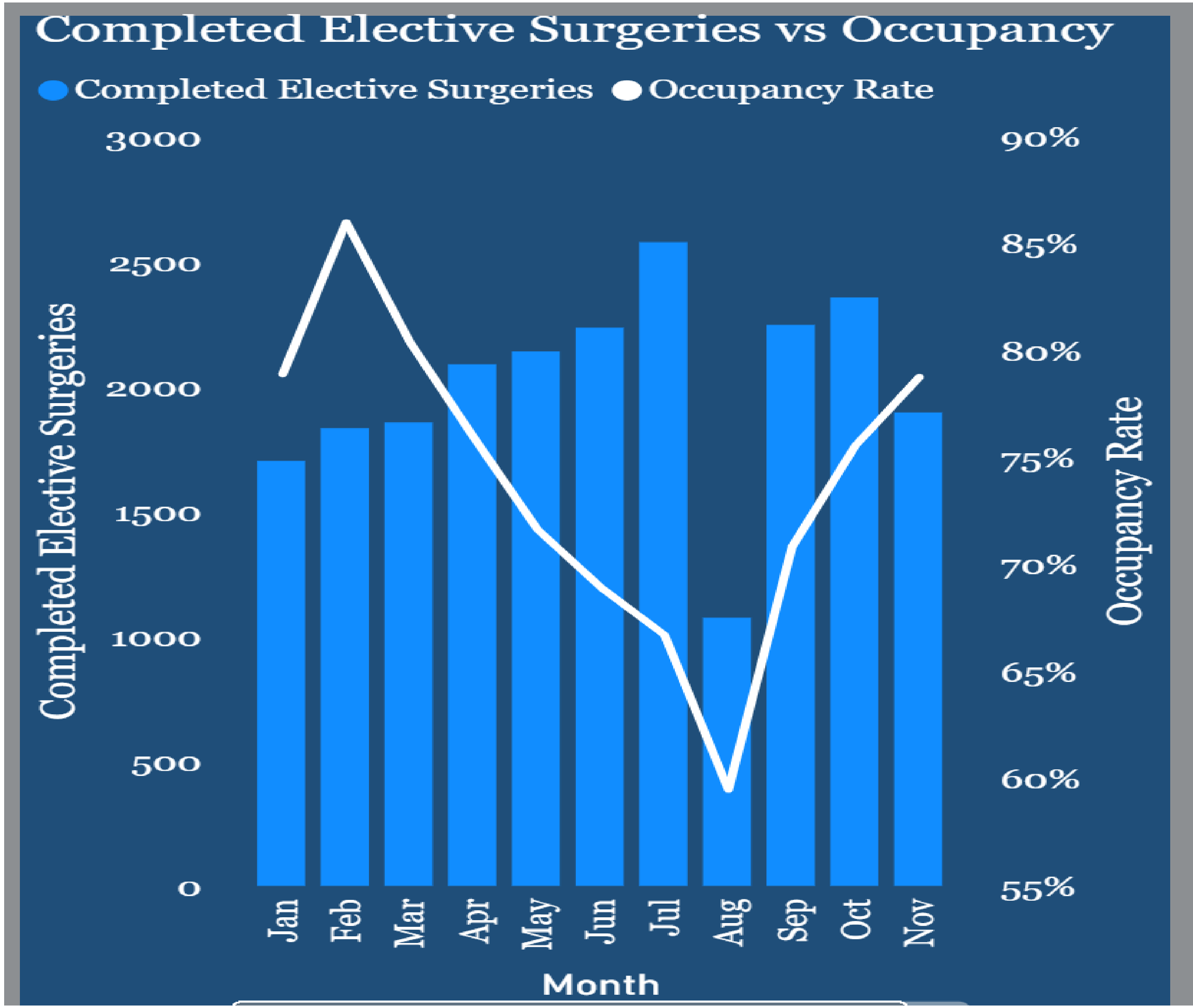
- Critical Care beds have the highest occupancy rate **(78%),** indicating they are the most heavily utilized and are often in high demand.

- Standard beds also have a high occupancy rate (**76%),** showing that they are consistently used across the hospital



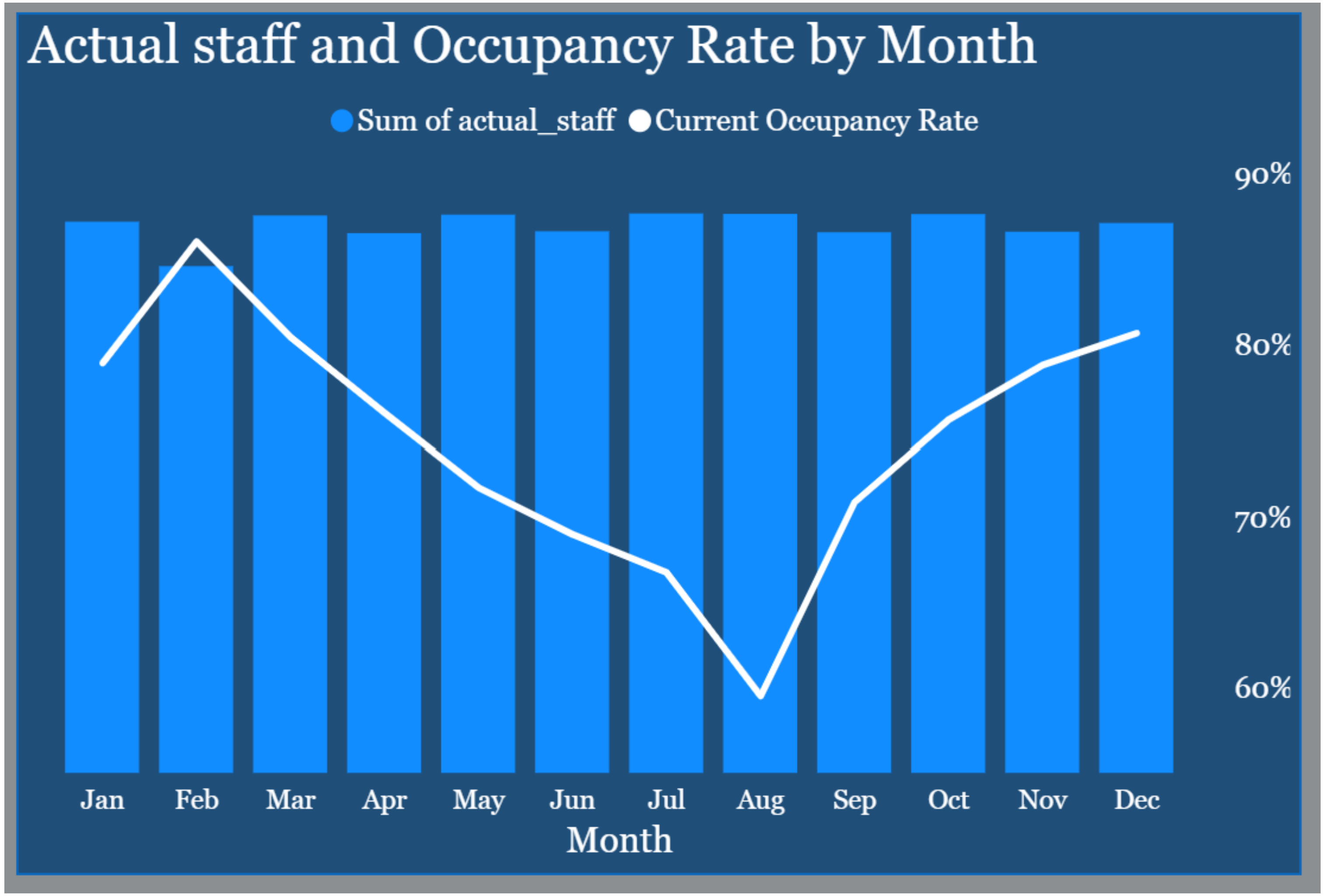
RELATIONSHIP BETWEEN ELECTIVE SURGERIES & OCCUPANCY RATE

Elective surgeries contribute to hospital workload; however, they are not the dominant driver of bed occupancy. The findings suggest that hospital capacity planning should consider multiple demand drivers rather than relying solely on elective surgery

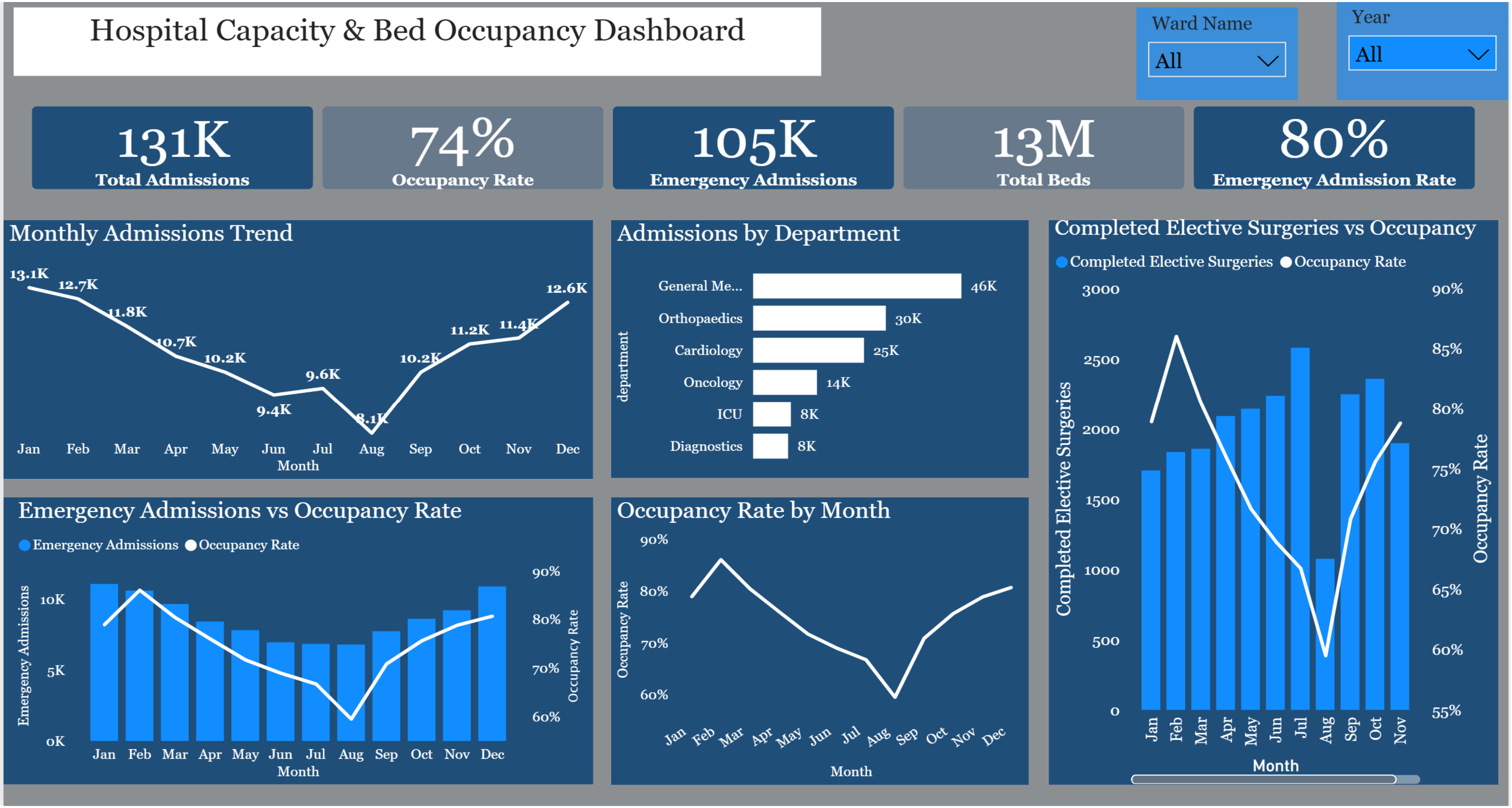


STAFFING AND OCCUPANCY
INSIGHT

- Although staffing level remain relatively stable through the year, they do not always align with occupancy rate



ALBION CARE EXECUTIVE DASHBOARD



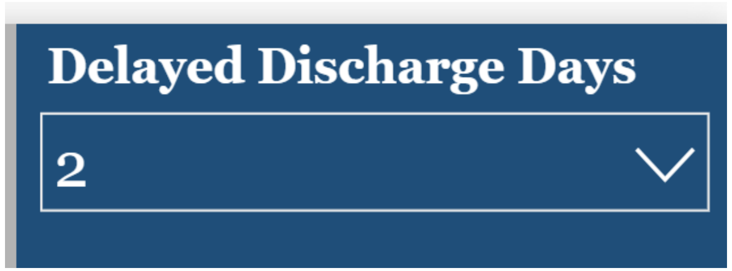
Delayed Discharge Scenario Analysis

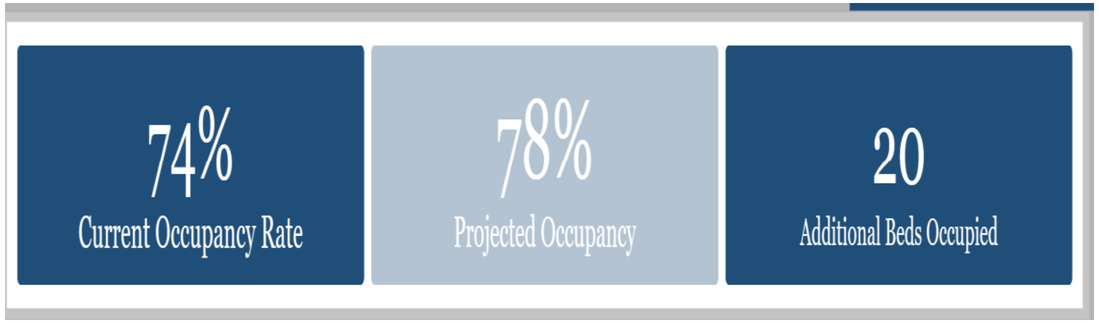
Scenario

A **2 days delay discharge** scenario was simulated to assess the impact of patient remaining in the hospital for additional **two days** after they were medically fit for discharge

Key Findings

Current Occupancy Rate **74%**
Projected Occupancy **78%**
Additional Beds Occupied **20**

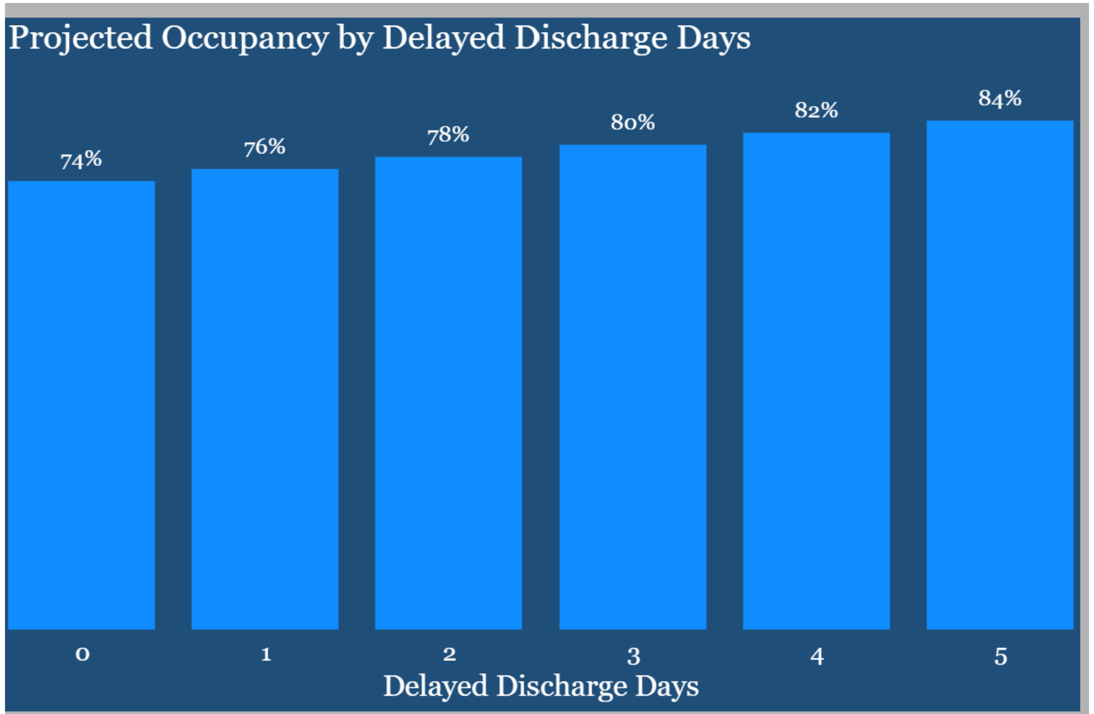


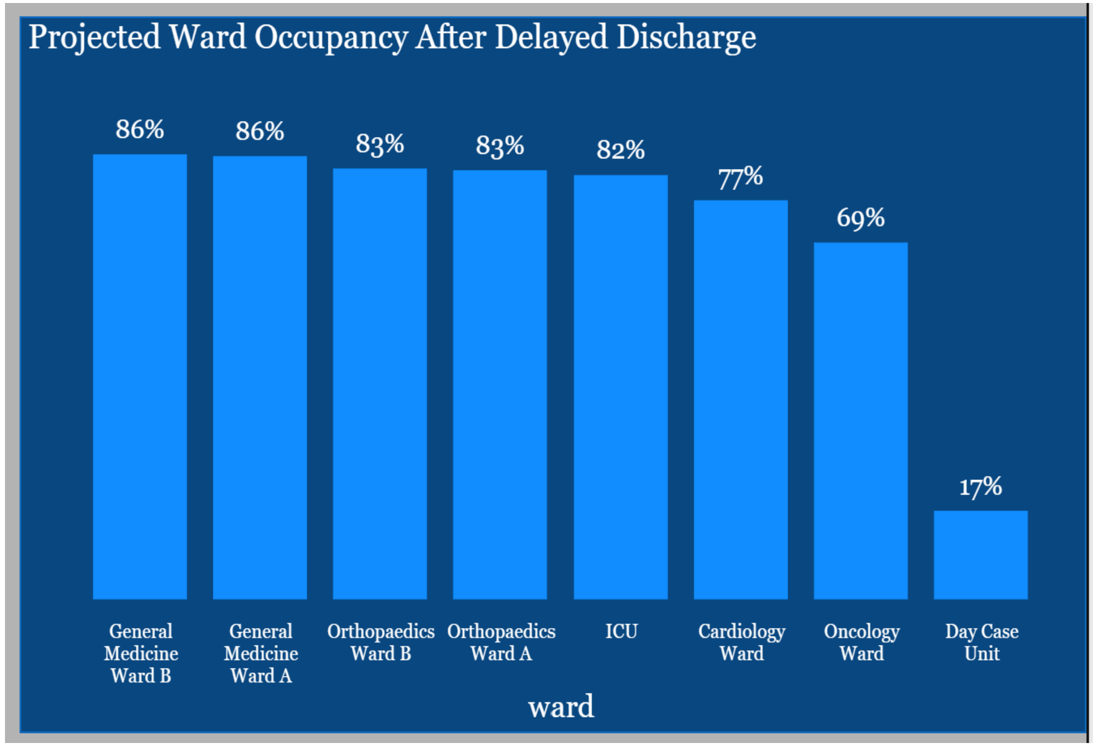


Delayed Discharge Analysis Results

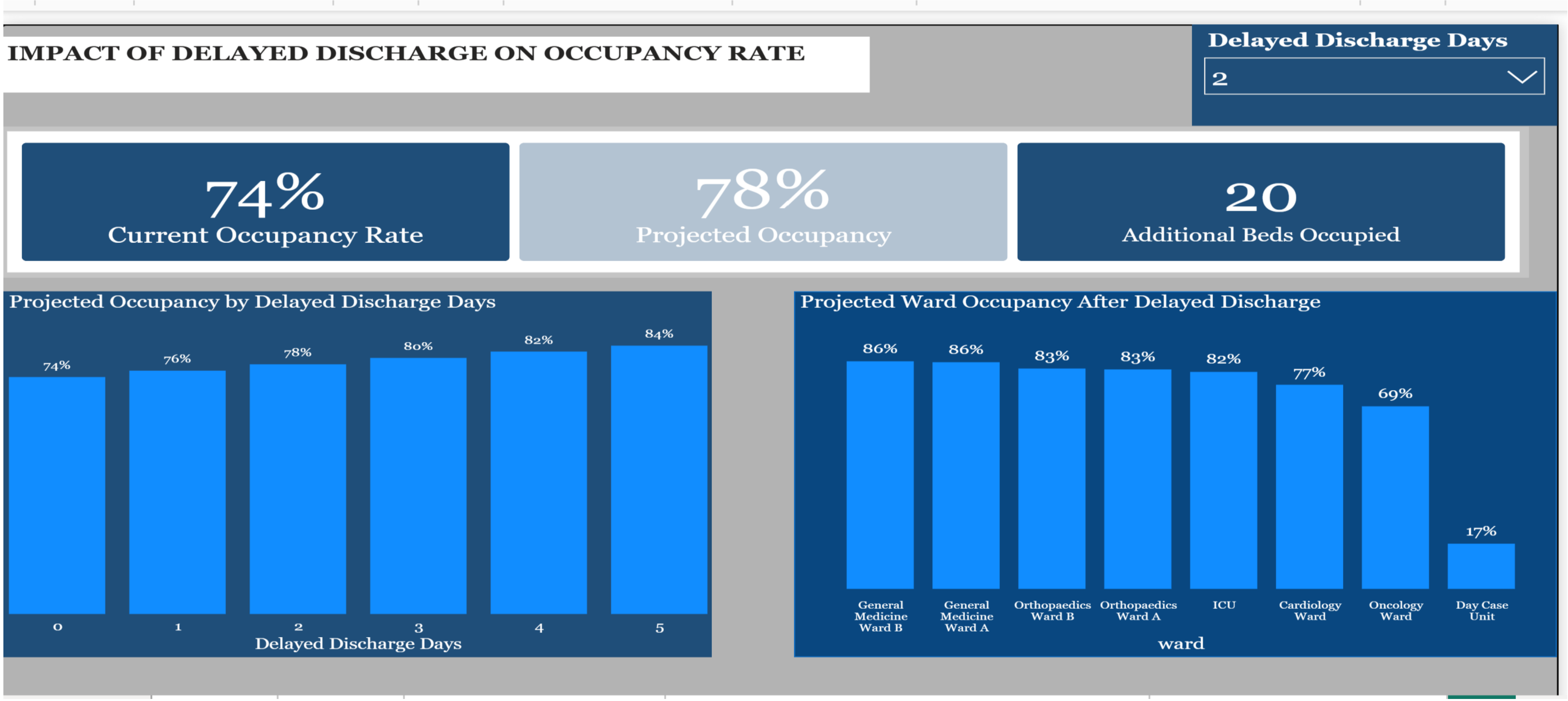
This chart shows that projected occupancy increases steadily as delayed discharge days increase. Longer discharge delays reduce bed availability and place greater pressure on hospital capacity.

General Medicine and Orthopedics record the highest projected occupancy following discharge delays, while the Day Case Unit remains the least affected. This helps identify which wards should be prioritized for discharge planning and resource allocation.



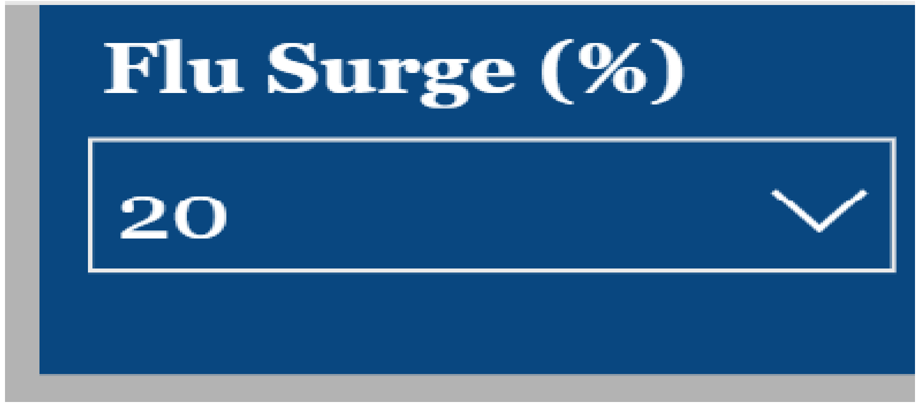


WHAT IF ANALYSIS DASHBOARD –DELAYED DISCHARGE



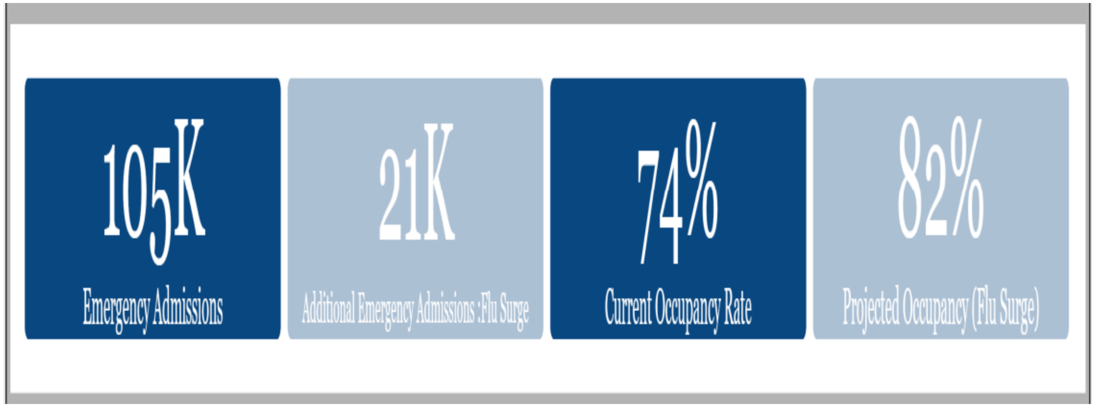
WHAT IF ANALYSIS DASHBOARD-FLU SURGE

A 20% flu surge scenario was simulated to assess the impact of increased emergency admissions on hospital occupancy and bed capacity.



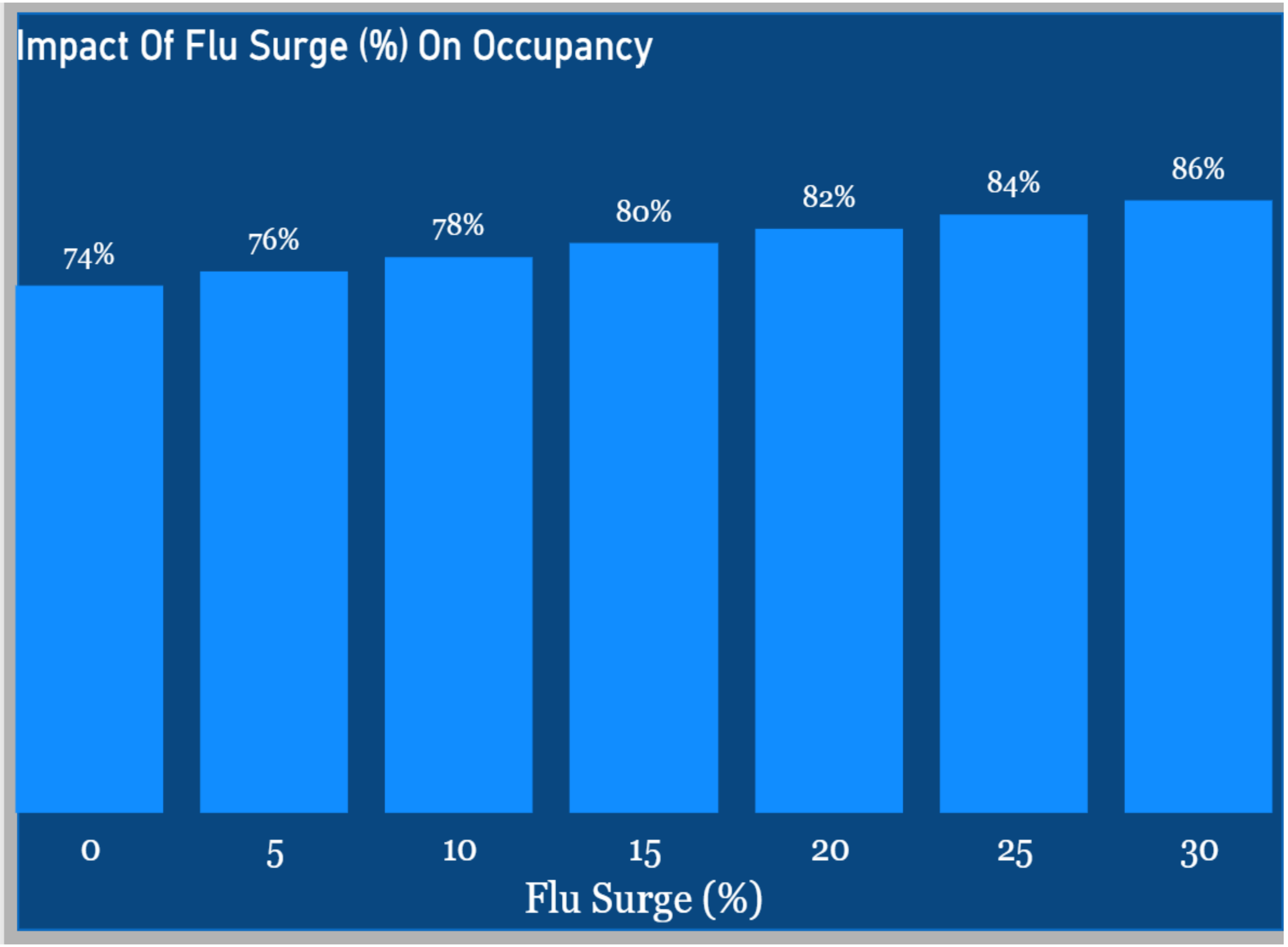
Emergency Admissions: 105K

- Additional Emergency Admissions: 21K
- * Current Occupancy Rate: 74%
- * Projected Occupancy: 82%



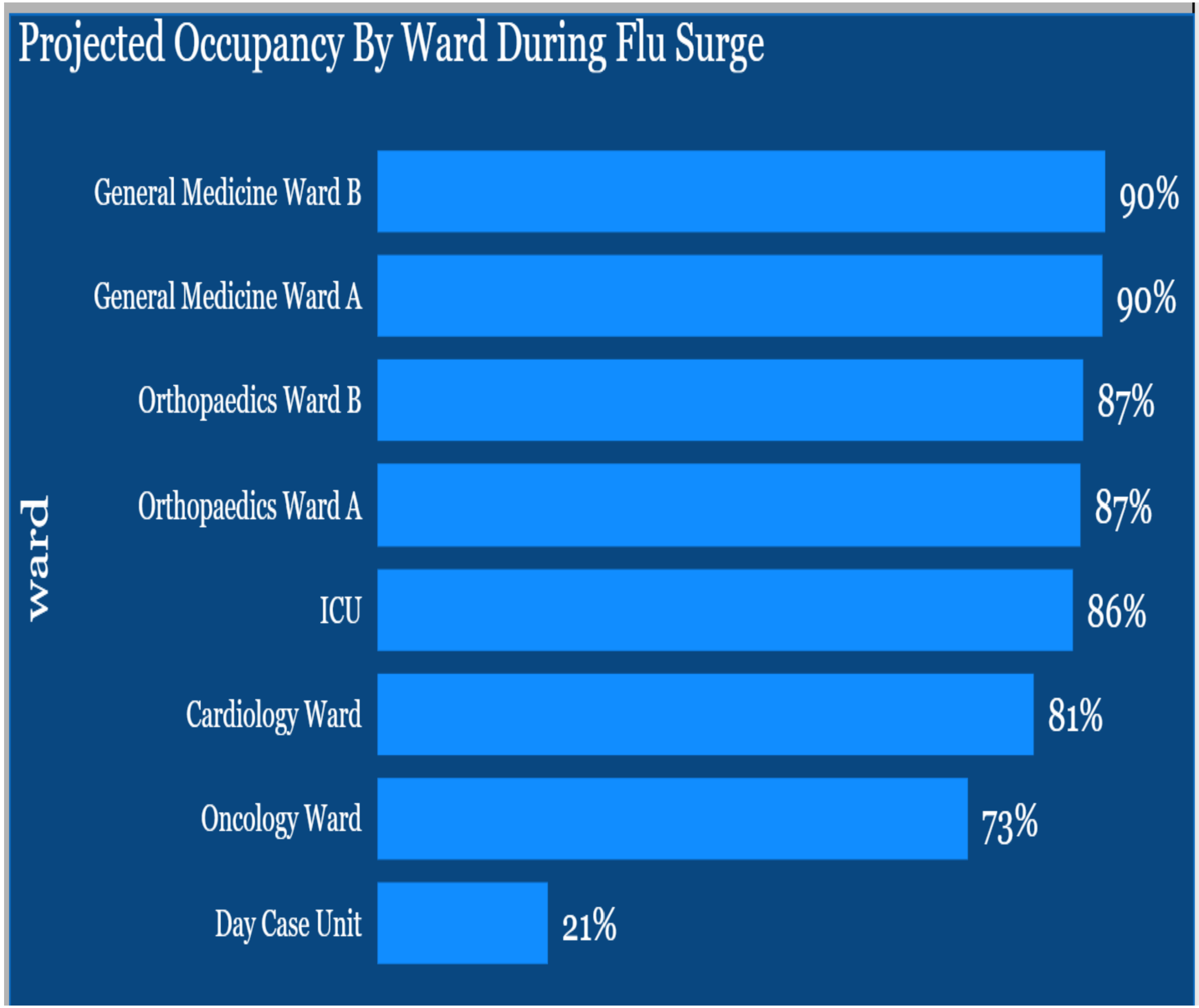
Flu Surge Analysis Results

This chart shows that occupancy increases gradually as flu-related admissions increase. The hospital would experience greater pressure on bed capacity during periods of high seasonal demand, highlighting the importance of proactive capacity planning.

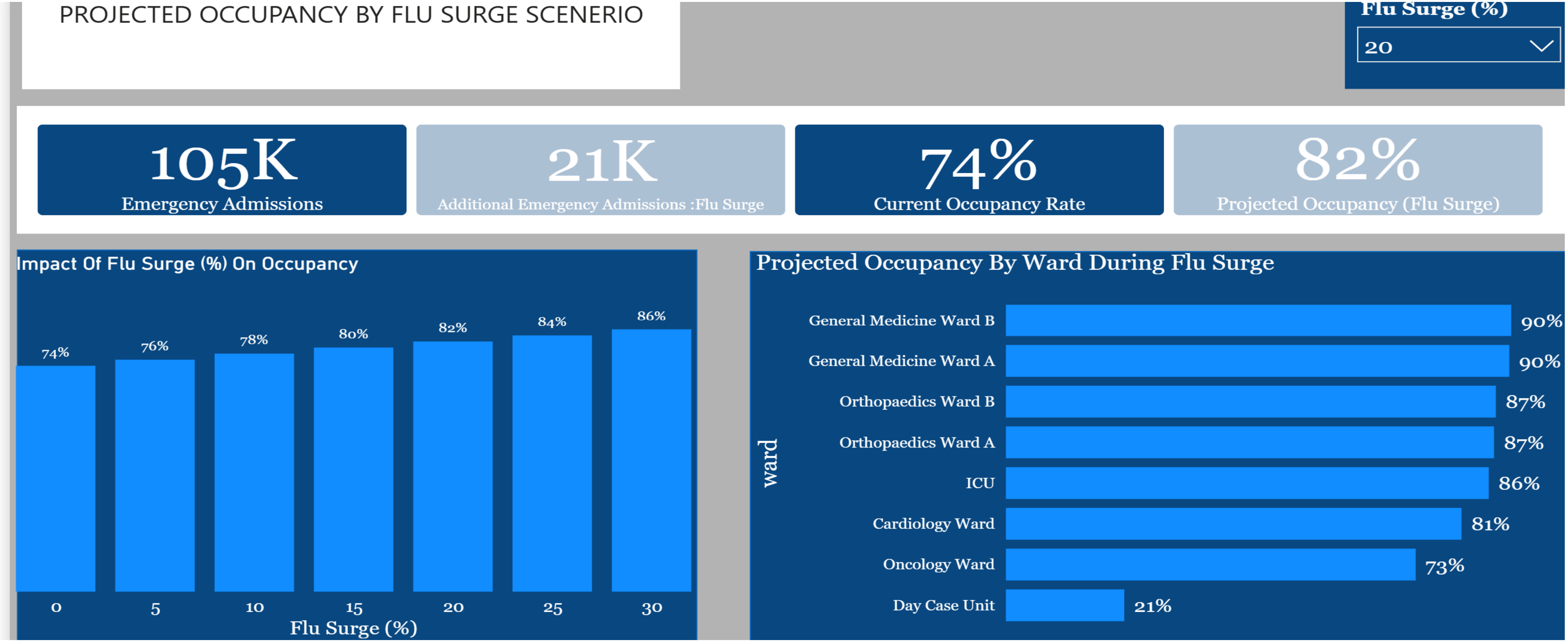


Impact of flu Surge on the Ward

This chart shows the wards most affected during a flu surge. General Medicine and Orthopedics reach the highest projected occupancy at around 90%, followed closely by the ICU. These findings suggest that hospital managers should prioritize these wards for additional staffing, bed capacity, and resource allocation during periods of increased flu-related admissions.



WHAT IF ANALYSIS DASHBOARD- FLU SURGE



RECOMMENDATIONS

- Strengthen emergency admission department capacity during flu season
- Improve patient discharge planning to optimize bed utilization
- Allocate more resources to high-demand departments
- Develop seasonal capacity planning strategies
- Investigate additional factors influencing occupancy
- **Increase staffing** during periods of high occupancy to ensure adequate patient care and reduce pressure on health care staff.
- Prioritize capacity planning for **Critical care** and **standard beds** and review utilization of **Day Care Beds**

CONCLUSION

Hospital management should adopt a data-driven approach to capacity planning by continuously monitoring admissions, occupancy trends, emergency demand, and departmental workloads. Using predictive analytics and timely resource allocation will improve bed utilization, reduce operational pressure, and enhance patient care.

THANK YOU